

SENG1050

Lab 2

In this lab, you will gain experience in constructing web pages by creating a HTML template and learning the usage of HTML tags.

[Task 1*](#) - [Task 2*](#) - [Task 3*](#) - [Task 4*](#)

Task 1 - Create a HTML Template

As all HTML pages are required to have a certain minimum structure, it is a good idea to keep a template file containing that structure. You can then copy the file each time you need to create a new web page.

1. Create a new directory. Perhaps you could call it `seng1050`.
If you're using a lab computer, it is recommended that you create a temporary folder in My Documents, which you can copy to your student Onedrive at the end of the lab.
2. Launch your favourite plain-text editor. This could be notepad, notepad++, or some other editor designed to produce `.txt` files.
Do not use a "word processor", such as Microsoft Word or a web authoring tool like Frontpage and Dreamweaver. The special formatting information used by word processors will make your HTML files **invalid**.
3. Create a new file in your text editor and cut-and-paste the following line:

```
<!DOCTYPE html>
```

This tells the Web browser that you are using HTML.

4. Below the first line, cut-and-paste the following lines:

```
<html>
<head>
<meta charset="UTF-8">
</head>

<body>
</body>

</html>
```

You now have the basic structure of a HTML document.

5. Next, create a title (`<title>`) element inside the head (`<head>`) element, resulting in the following:

```
<head>
<title>Put the Title Here</title>
<meta charset="UTF-8">
</head>
```

You should also add a level-1 heading (`<h1>`) element inside the body (`<body>`) element, resulting in the following:

```
<body>
<h1>Put the Most Important Heading Here</h1>
</body>
```

6. Save the file you have created into the new directory in step 1. Call the file `template.html`.

Open the file in your favourite Web browser (either start the Web browser and use the `File` menu, or simply double-click on the file). It should look something like this. (The title bar might look different on different browsers or OSes.)

1: Put the Title Here

Put the Most Important Heading Here

Be careful to ensure the file is saved as `template.html` and not `template.html.txt` - some text editors will do the latter by default when creating a new file. To avoid this, be sure to set the `File Type` to all files.

It is also a good idea to slightly change the settings on Microsoft's File Explorer.

From the menubar, select `Tools` then `Folder Options`. Select `View`, and make sure the box labelled `Hide extensions for known file types` is **not** ticked.

This will enable you to see full filenames, including whether the suffix is `.html` or `.html.txt`.

Task 2 - Adding HTML Tags to Your Content

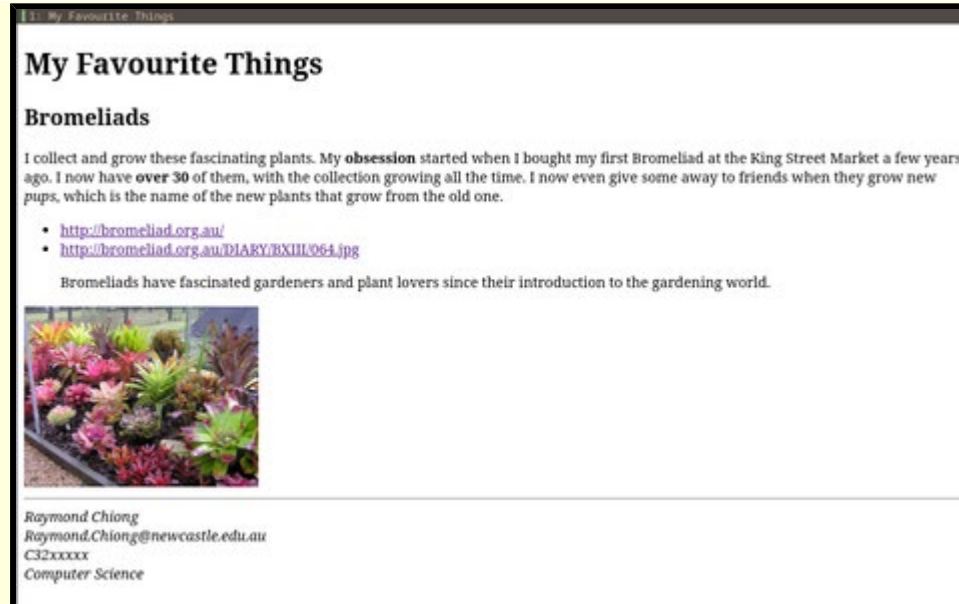
1. Before you start, think about three of your favourite things.
For each thing, find a webpage and an image related to it.
Copy the original URLs of both the webpage and the image, and download the image to your computer, you'll need these later.
2. Make a copy of `template.html` and call it `index.html`.
Open `index.html` in your favourite plain-text editor.
3. Change the content of the title (`<title>`) and level-1 heading (`<h1>`) elements to be `My Favourite Things`.
4. Immediately after the level-1 heading element, create three level-2 heading (`<h2>`) elements.
Each level-2 heading element will contain one of your favourite things as the content.
5. Add paragraph tags (`<p>`) around each paragraph of text describing a favourite thing.
Use `<strong>` and `<em>` tags to strongly emphasize some phrase or words in your description.
6. Under each paragraph, start an unordered list (`<ul>`) element.
Each list should contain two items (`<li>`): the URL of the Web page about that favourite thing, and the *original* URL of the image related to that favourite thing.
Each URL should be hyperlinked to its destination, i.e., using `<a href="url">url</a>`.
7. After the unordered list, add a quotation related to your favourite item using the `<blockquote>` tag.
You should use the `'cite'` attribute to add a URL referring to the source of the citation.
8. After each blockquote, place an image (`<img>`) element linking to the local copy of the image you downloaded.
As an image element is an inline element and must be placed inside a block element, place a `<div>` tag around the `<img>` tag. For example:

```
<div>

</div>
```

9. At the bottom of the page, add a horizontal rule (`<hr>`), and then add an address (`<address>`) element.
Place your contact details (name, email address, student number and degree program) as content of the address (`<address>`) element.
Separate each logical line of the address with a break (`<br>`) tag.

Open the file in your favourite Web browser. It should look something like this.



Go to task 3 now, come back here at the end if you have time.

If you have time, experiment with some other tags introduced in Lecture 2. You might include

- The creation date of your html file using `<sup>` tag to format the date.
- A sample of verse (perhaps from the Internet) referring to your favourite things formatted using `<blockquote>`.
- A definition list.
- Abbreviations.
- Special Characters: `<`, `>` &.
- A sample of computer code (say Java or C++) with input and output.
- Any other use of tags from Lecture 2, such as input instructions, definitions or citations.

Task 3 - Learn to link your HTML documents

Linking to Identifiers

1. Create another copy of `index.html` and call it `index2.html`.
Open this new file in your favourite plain-text editor.
2. At the top of the page (under the `<h1>` element), create a `<div>` element.
3. Inside the `<div>` element, create three anchor (`<a>`) elements, one for each of your favourite things.
Insert unique names preceded by the `#` sign in the `href` attribute field of each anchor element.
4. Now, complete the linking by putting the chosen names in the corresponding `<h2>` elements' `id` attribute field.
5. Verify that your links are working correctly.
6. Finally, add one `<hr>` before and after the `<div>` element you created for links.

Linking among different files

1. Create one separate html file for each of your favourite things by copying the original `index.html`.
Each file will contain information about one of your favourite thing. You can reuse what you have starting from the `<h2>` heading to the image of that particular favourite thing.
Rename each of the files, e.g., as `favourite01.html`, `favourite02.html`, etc.
2. Create another new html file from the original `template.html`. Treat this as your menu page and name it as `favourite.html`.

Create a `<div>` element with links to each of your favourite things saved in different files.

3. Now, open each of the files containing your favourite things (`favourite01.html`, `favourite02.html`, etc.) and at the bottom, add a link to go back to the menu page (`favourite.html`).
 4. Visit each link to verify that those are working properly
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Task 4 - HTML Validation

As you should learn very quickly in this course, just because a Web page *looks* correct in *your* browser on *your* computer, doesn't mean that it is correct. The easiest way to check whether a web page is correct is to use a web validation service.

1. Go to the [W3C MarkUp Validation Service](#).
2. On the `Validate By File Upload` tab in the web page, select your web page file, then click on the `Check` button.

If all is well, then the validator will announce that your page contains no errors. If it does find any errors, then it will try to tell you why the error occurred and on which line of the file. Try to ensure your webpages pass validation with no errors OR warnings.

Tip: always fix the *first* error that the validator reports (the one nearest the top of the list), then revalidate (use the `Reload/Refresh` button of your browser. Fixing the first reported error will often fix the second as well - and the third, and the fourth ...

All code you write for your assignments MUST pass validation. If any errors are reported, you will lose a significant amount of marks.

For assignment 2, this will be stricter, even warnings will need to be addressed.
